BIOCHEMISTRY & CELL BIOLOGY

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Q1: Write on classification of carbohydrates and give an example for each type?

Carbohydrates are classified into three main types based on molecular complexity:

- **1. Monosaccharides:** Simplest sugars that cannot be hydrolyzed further. *Examples: Glucose, Fructose, Galactose, Ribose.*
- **2. Disaccharides:** Formed by condensation of two monosaccharides. *Examples: Sucrose (Glucose + Fructose), Lactose (Glucose + Galactose), Maltose (Glucose + Glucose).*
- **3. Polysaccharides:** Polymers consisting of three or more sugar units. *Examples: Glycogen, Starch, Cellulose.*

Q2: Compare between Prokaryotic and Eukaryotic Cells?

Aspect	Prokaryotic Cells	Eukaryotic Cells
Size	Smaller than eukaryotic cells	Larger in size
Complexity	More simple structure	More complex structure
Nucleus	Lack membrane-bound nucleus	Have true membrane-bound nucleus
DNA Structure	Circular and directly in cytoplasm; without proteins	Linear and associated with proteins (histones)
Cell Wall	Present on outer surface	Present in plants & fungi; absent in animals
Cell Multiplication	Multiply at a faster rate (Binary Fission)	Multiply at a slower rate (Mitosis/Meiosis)
Examples	Bacteria, blue-green algae	Plants, fungi, animals

Q3: Compare between Germ Cells and Somatic Cells?

Feature	Germ Cells (Gametes)	Somatic Cells
Definition	Reproductive sex cells involved in sexual reproduction	All biological body cells forming tissues/organs
Chromosome Sets	One set of chromosomes (Haploid - N)	Two sets of chromosomes (Diploid - 2N)
Number of Chromosomes	23 single chromosomes (in humans)	46 chromosomes or 23 homologous pairs
Examples	Human Egg (Ovum), Human Sperm (Spermatozoa)	Neuron, Epithelial cells, Blood cells, Muscle cells

Q4: Describe the structure and functions of Cell Nucleus, Cell Membrane, Mitochondria, and ER

A) Cell Nucleus (The Control Center)

- **Structure:** Surrounded by a double nuclear envelope/membrane with pores; contains genetic material (DNA) and the nucleolus.
- **Functions:** Stores genetic information; acts as the control center of cellular activities; responsible for DNA replication and gene transcription; produces ribosomes via the nucleolus.

B) Cell Membrane (Plasma Membrane)

- **Structure:** Mainly composed of a phospholipid bilayer embedded with functional proteins. Phospholipids possess hydrophilic heads facing aqueous environments (outer/inner) and hydrophobic tails oriented inward.
- **Functions:** Isolates cell contents from external environment; provides physical protection; selectively regulates nutrient uptake and toxic waste removal; mediates cell-to-cell communication.

C) Mitochondria (The Powerhouse of the Cell)

- **Structure:** Oval double-membraned organelle; outer membrane is smooth, inner membrane is highly folded into cristae. Encloses an internal matrix containing circular DNA, RNA, and ribosomes.
- **Functions:** Primary site of cellular respiration and ATP generation (energy production).

D) Endoplasmic Reticulum (ER)

- **Structure:** A vast interconnected network of tubular and flattened sacs located adjacent to the nucleus, divided into Rough ER (studded with ribosomes) and Smooth ER (lacks ribosomes).
- **Functions:** Rough ER (RER): synthesizes and processes proteins. Smooth ER (SER): primary site for lipid and steroid synthesis and drug detoxification.

Q5: Compare between Isotonic, Hypotonic, and Hypertonic Solutions?

Solution Type	Solute Concentration	Water Movement	Effect on Cell Size
Isotonic	Equal inside and outside cell	No net water movement (dynamic equilibrium)	Cell size remains constant & unchanged
Hypotonic	Outside solute is lower (dilute)	Water enters cell (Endosmosis)	Cell swells and may burst / lyse (turgid in plants)
Hypertonic	Outside solute is higher (concentrated)	Water leaves cell (Exosmosis)	Cell shrinks and shrivels (plasmolysis in plants)

Q6: Write on Types of Porters (Transporters)? [2.5 Marks]

- **Uniporter:** Transports a single type of molecule across the membrane in one direction.
- **Symporter (Cotransporter):** Transports two different molecules across the membrane simultaneously in the same direction.
- **Antiporter (Exchanger):** Transports two different molecules across the membrane simultaneously in opposite directions.

Q7: Write about the different types of Osmosis? [2.5 Marks]

- **1. Endosmosis:** The inward movement of water into a cell when placed in a hypotonic solution (solute concentration outside is lower than inside). This causes the cell to swell.
- **2. Exosmosis:** The outward movement of water from a cell when placed in a hypertonic solution (solute concentration outside is higher than inside). This causes the cell to shrink/shrivel.

Q8: What are the steps of Binary Fission?

Binary fission is the simplest and most common form of asexual reproduction in bacteria under favorable conditions, producing two identical daughter cells:

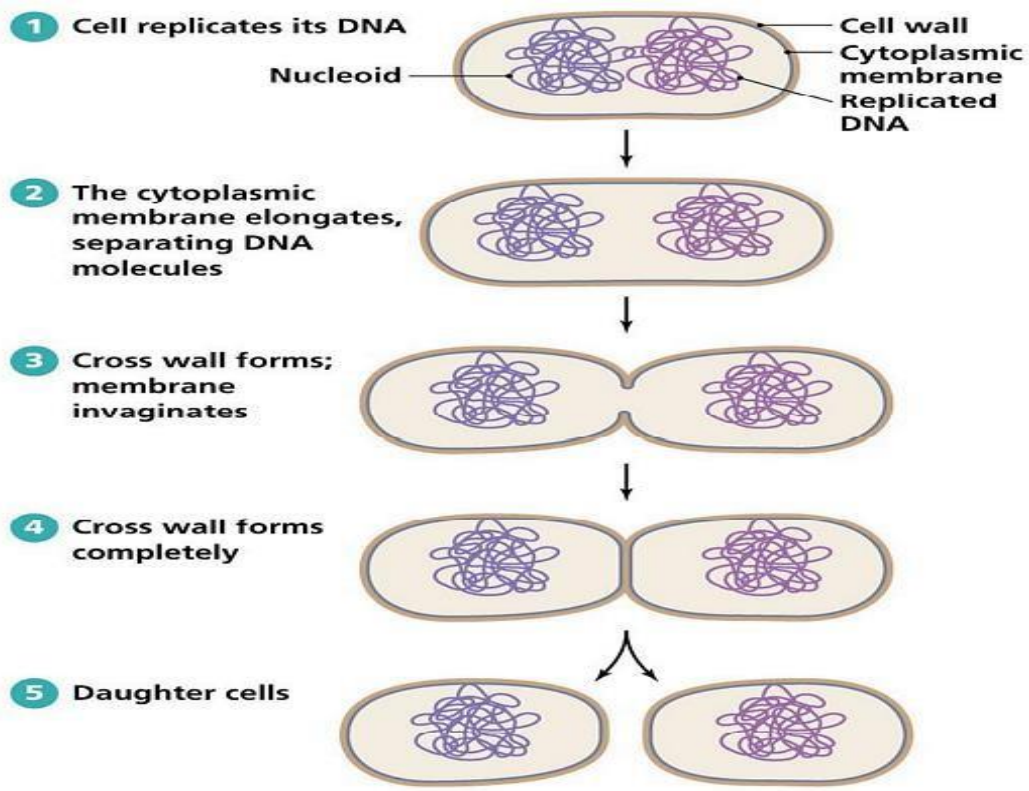
Step 1: The parent bacterial cell increases in size and elongates.

Step 2: Circular bacterial DNA replicates, forming two identical copies.

Step 3: The two DNA molecules migrate towards opposite poles of the cell.

Step 4: The cell membrane and peptidoglycan cell wall begin to constrict inward at the center.

Step 5: A transverse septum forms completely, partitioning the cell into two genetically identical daughter cells.



Q9: Explain the Sodium-Potassium Pump (Na^+/K^+ ATPase) mechanism in living cells

The primary active transport mechanism operates in a cyclical sequence:

Step 1: Cytosolic Binding - The pump conformation opens toward the cell interior, and 3 Na^+ ions bind to high-affinity binding sites.

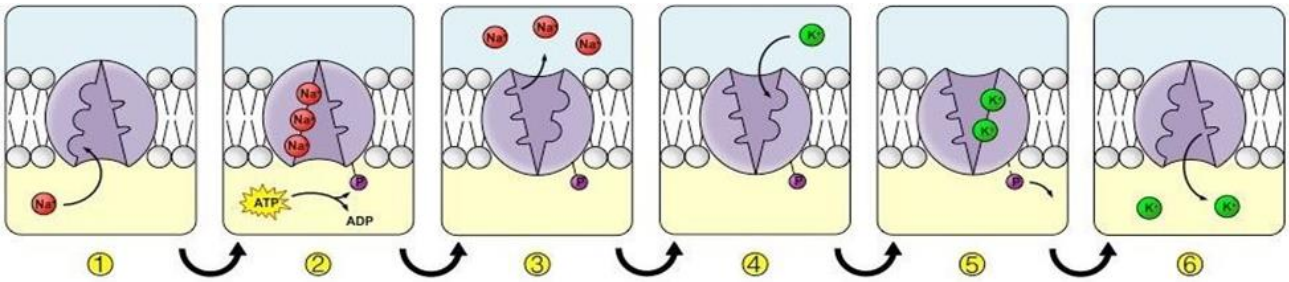
Step 2: Phosphorylation - ATP is hydrolyzed to ADP and inorganic phosphate; the terminal phosphate group is covalently attached to the pump (phosphorylation).

Step 3: Conformational Change & Na^+ Release - Phosphorylation triggers a structural change, opening the pump to the extracellular space and releasing the 3 Na^+ ions.

Step 4: Extracellular K^+ Binding - 2 K^+ ions from the extracellular fluid bind to high-affinity outer sites of the transporter.

Step 5: Dephosphorylation - The bound phosphate group is released from the pump protein (dephosphorylation).

Step 6: Return Conformation & K^+ Release - The pump reverts to its original inward-facing conformation, releasing the 2 K^+ ions into the cytosol to repeat the cycle.



Q10: Describe the types of Bulk Transport

Bulk transport moves macromolecules and large particles across membranes via membrane-bound vesicles (requires energy):

- **1. Endocytosis:** Uptake of large macromolecules into the cell through vesicle invagination:
  - **• Phagocytosis ('Cell Eating'):** Engulfment of large solid particles (e.g., bacteria, cellular debris).
  - **• Pinocytosis ('Cell Drinking'):** Non-specific internalization of extracellular fluid and dissolved solutes.
- **2. Exocytosis:** Secretion or release of intracellular substances into the extracellular space as secretory vesicles fuse with the plasma membrane.

Q11: Discuss the Three Types of Passive Transport

Passive transport is the movement of substances down their concentration gradient without cellular energy expenditure (ATP):

1. Simple Diffusion: Direct, spontaneous net movement of small, uncharged molecules across the lipid bilayer down their gradient. Examples: Gases (O2, CO2), lipids, and small nonpolar molecules.

2. Facilitated Diffusion: Transport of polar or charged molecules (glucose, amino acids, ions) mediated by specialized transmembrane proteins:

a. Channel Proteins: Form hydrophilic water-filled transmembrane pores allowing rapid selective diffusion of specific ions/solutes.

b. Carrier Proteins: Undergo subtle conformational shifts upon solute binding to translocate the molecule across the lipid bilayer.

3. Osmosis: The specialized net diffusion of water molecules across a selectively permeable membrane from an area of higher water potential (lower solute concentration) to lower water potential (higher solute concentration).